Chapter 15
Evaluation: Inspections, Analytics, and Models

Objectives

The main aims of this chapter are to:

- Describe the key concepts associated with inspection methods.
- Explain how to do heuristic evaluation and walkthroughs.
- Explain the role of analytics in evaluation.
- Describe how to use Fitts' Law – a predictive model.

15.1 Introduction

The evaluation methods described so far in this book have involved interaction with, or direct observation of, users. In this chapter we introduce methods that are based on understanding users through knowledge codified in heuristics, or data collected remotely, or models that predict users' performance. None of these methods requires users to be present during the evaluation. Inspection methods typically involve an expert role-playing the users for whom the product is designed, analyzing aspects of an interface, and identifying any potential usability problems by using a set of guidelines. The most well known are heuristic evaluation and walkthroughs. Analytics involves user interaction logging, which is usually done remotely. Predictive models involve analyzing the various physical and mental operations that are needed to perform particular tasks at the interface and operationalizing them as quantitative measures. One of the most commonly used predictive models is Fitts' Law.

15.2 Inspections: Heuristic Evaluation and Walkthroughs

Sometimes users are not easily accessible, or involving them is too expensive or it takes too long. In such circumstances other people, usually referred to as experts, can provide feedback. These are people who are knowledgeable about both interaction design and the needs and typical behavior of users. Various inspection methods were developed as alternatives to usability testing in the early 1990s, drawing on software engineering practice where code and other types of inspections are commonly used. Inspection methods for interaction design include heuristic evaluations, and walkthroughs, in which experts examine the interface of an interactive product, often role-playing typical users, and suggest problems users would likely have when interacting with it. One of the attractions of these methods is that they can be used at any stage of a design project. They can also be used to complement user testing.

15.2.1 Heuristic Evaluation

Heuristic evaluation is a usability inspection method that was developed by Nielsen and his colleagues (Nielsen and Mohlich, 1990; Nielsen, 1994a) and others (Hollingshead and Novick, 2007), and later modified by other researchers for evaluating specific types of systems (e.g. Mankoff *et al*, 2003; Pinelle *et al*, 2009). In heuristic evaluation, experts, guided by a set of usability principles known as heuristics, evaluate whether user-interface elements, such as dialog boxes, menus, navigation structure, online help, and so on, conform to tried and tested principles. These heuristics closely resemble high-level design principles (e.g. making designs consistent, reducing memory load, and using terms that users understand). The original set of heuristics for HCI evaluation was developed by Jakob Nielsen and his colleagues who derived them empirically from an analysis of 249 usability problems (Nielsen, 1994b); a revised version of these heuristics is listed below (Nielsen, 2014: useit.com):

- **Visibility of system status**
  The system should always keep users informed about what is going on, through appropriate feedback within reasonable time.
- **Match between system and the real world**
  The system should speak the users' language, with words, phrases, and concepts familiar to the user, rather than system-oriented terms. Follow real-world conventions, making information appear in a natural and logical order.
- **User control and freedom**
  Users often choose system functions by mistake and will need a clearly marked emergency exit to leave the unwanted state without having to go through an extended dialog. Support undo and redo.
- **Consistency and standards**
  Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform conventions.
- **Error prevention**
  Even better than good error messages is a careful design that prevents a problem from occurring in the first place. Either eliminate error-prone conditions or check for them and present users with a confirmation option before they commit to the action.
- **Recognition rather than recall**
  Minimize the user's memory load by making objects, actions, and options visible. The user should not have to remember information from one part of the dialog to another. Instructions for use of the system should be visible or easily retrievable whenever appropriate.
- **Flexibility and efficiency of use**
  Accelerators – unseen by the novice user – may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.

- **Aesthetic and minimalist design**
  Dialogs should not contain information that is irrelevant or rarely needed. Every extra unit of information in a dialog competes with the relevant units of information and diminishes their relative visibility.
- **Help users recognize, diagnose, and recover from errors**
  Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.
- **Help and documentation**
  Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large. Before reading on, watch the YouTube video developed by David Lazarus.

> Video developed by David Lazarus, which aims to provide insight into Jakob Nielsen's 10 Usability Heuristics for Interface Design, is available at http://youtu.be/hWc0Fd2AS3s

These heuristics are intended to be used by judging aspects of the interface against them. For example, if a new social networking system is being evaluated, the evaluator might consider how a user would find out how to add friends to her network. The evaluator is meant to go through the interface several times, inspecting the various interaction elements and comparing them with the list of usability principles, i.e. the heuristics. At each iteration, usability problems will be identified or their diagnosis will be refined, until she is satisfied that the majority of them are clear.

Although many heuristics apply to most products (e.g. be consistent and provide meaningful feedback), some of the core heuristics are too general for evaluating products that have come onto the market since Nielsen and Mohlich first developed the method, such as smartphones and other mobile devices, digital toys, online communities, ambient devices, and new web services. Nielsen suggests developing category-specific heuristics that apply to a specific class of product as a supplement to the general heuristics. Evaluators and researchers have therefore typically developed their own heuristics by tailoring Nielsen's heuristics with other design guidelines, market research, and requirements documents. Exactly which heuristics are appropriate and how many are needed for different products is debatable and depends on the goals of the evaluation, but most sets of heuristics have between five and ten items. This number provides a good range of usability criteria by which to judge the various aspects of an interface. More than ten becomes difficult for evaluators to remember; fewer than five tends not to be sufficiently discriminating.

A key question that is frequently asked is how many evaluators are needed to carry out a thorough heuristic evaluation. While one evaluator can identify a large number of problems, she may not catch all of them. She may also have a tendency to concentrate more on one aspect at the expense of missing others. For example, in a study of heuristic evaluation where 19 evaluators were asked to find 16 usability problems in a voice response system allowing customers access to their bank accounts, Nielsen (1992) found a substantial difference between the number and type of usability problems found by the different evaluators. He also notes that while some usability problems are very easy to find by all evaluators, there are some problems that are found by very few experts. Therefore, he argues that it is important to involve multiple evaluators in any heuristic evaluation and recommends between three and five evaluators. His findings suggest that they can typically identify around 75% of the total usability problems, as shown in Figure 15.1 (Nielsen, 1994a).

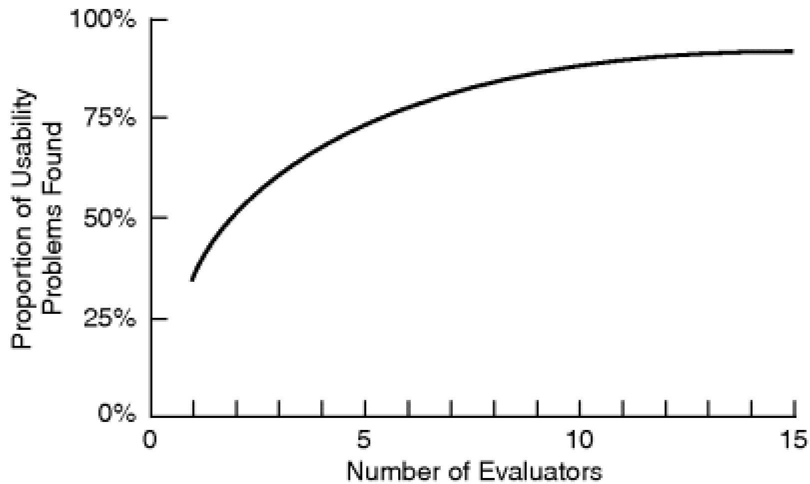

Figure 15.1 Curve showing the proportion of usability problems in an interface found by heuristic evaluation using various numbers of evaluators. The curve represents the average of six case studies of heuristic evaluation

Source: **Usability Inspection Methods, J. Nielson & R.L. Mack ©1994. Reproduced with permission of John Wiley & Sons Inc.**

However, employing multiple experts can be costly. Skillful experts can capture many of the usability problems by themselves and some consultancies now use this technique as the basis for critiquing interactive products – a process that has become known as an expert critique or expert crit in some countries. But using only one or two experts to conduct a heuristic evaluation can be problematic since research has challenged Nielsen's findings and questioned whether even three to five evaluators is adequate. For example, Cockton and Woolrych (2001) and Woolrych and Cockton (2001) point out that the number of experts needed to find 75% of problems depends on the nature of the problems. Their analysis of problem frequency and severity suggests that highly misleading findings can result.

The conclusion from this is that more is better, but more is also more expensive. However, because users and special facilities are not needed for heuristic evaluation and it is comparatively inexpensive and quick, it is popular with developers and is often known as discount evaluation. For a quick evaluation of an early design, one or two experts can probably identify most potential usability problems, but if a thorough evaluation of a fully working prototype is needed then having a team of experts conducting the evaluation and comparing their findings would be advisable.

Heuristic Evaluation for Websites

In recent years, considerable attention has been focused on the web, mobile apps, and other digital technologies. Heuristics for evaluating websites have become increasingly important and several slightly different sets of heuristics have been developed. Box 15.1 contains an extract from a version compiled by web developer Andy Budd that places a stronger emphasis on information content than Nielsen's original heuristics.

BOX 15.1

Extract from the Heuristics Developed by Budd (2007) that Emphasize Web Design Issues

Clarity

Make the system as clear, concise, and meaningful as possible for the intended audience.

- Write clear, concise copy
- Only use technical language for a technical audience
- Write clear and meaningful labels
- Use meaningful icons.

Minimize unnecessary complexity and cognitive load

Make the system as simple as possible for users to accomplish their tasks.

- Remove unnecessary functionality, process steps, and visual clutter
- Use progressive disclosure to hide advanced features
- Break down complicated processes into multiple steps
- Prioritize using size, shape, color, alignment, and proximity.

Provide users with context

Interfaces should provide users with a sense of context in time and space.

- Provide a clear site name and purpose
- Highlight the current section in the navigation
- Provide a breadcrumb trail

- Use appropriate feedback messages
- Show number of steps in a process
- Reduce perception of latency by providing visual cues (e.g. progress indicator) or by allowing users to complete other tasks while waiting.

Promote a pleasurable and positive user experience

The user should be treated with respect and the design should be aesthetically pleasing and promote a pleasurable and rewarding experience.

- Create a pleasurable and attractive design
- Provide easily attainable goals
- Provide rewards for usage and progression. ■

Activity 15.1

1. Select a website that you regularly visit and evaluate it using the heuristics in Box 15.1. Do these heuristics help you to identify important usability and user experience issues?
2. Does being aware of the heuristics influence how you interact with the website in any way?

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A similar approach to Budd's is also taken by Leigh Howells in her web article entitled "A guide to heuristic website reviews" (Howells, 2011). Howells' approach goes one step further and suggests a way of quantifying the heuristics and consequently also the heuristic evaluation, which can then be presented using various types of visual diagrams.

Turning Design Guidelines into Heuristics

There is a strong relationship between design guidelines and the heuristics used in heuristic evaluation. So another approach to developing heuristics for evaluating the many different types of systems now available – mobile, tabletop, online communities, ambient, etc. – is to convert design guidelines into heuristics for evaluation. As a first step, evaluators sometimes translate design guidelines into questions for use in heuristic evaluation. This practice has become quite widespread for addressing usability and user experience concerns for specific types of interactive product. For example Väänänen-Vainio-Mattila and Waljas (2009) from the University of Tempere in Finland took this approach when developing heuristics for a web service user experience. They tried to identify what they called 'hedonic heuristics,' which was a new kind of heuristic that directly addressed how users feel about their interactions. These were based on design guidelines concerning whether the user feels that the web service provides a lively place where it is enjoyable to spend time, and whether it satisfies the user's curiosity by frequently offering interesting content. When stated as questions these become: Is the service a lively place where it is enjoyable to spend time? Does the service satisfy users' curiosity by frequently offering interesting content?

Activity 15.2

Consider the following design guidelines for information design and for each one suggest a question that could be used in heuristic evaluation:

1. Good graphical design is important. Reading long sentences, paragraphs, and documents is difficult on screen, so break material into discrete, meaningful chunks to give the website structure (Horton, 2005).
2. Avoid excessive use of color. Color is useful for indicating different kinds of information, i.e. cueing (Koyani *et al*, 2004).
3. Avoid gratuitous use of graphics and animation. In addition to increasing download time, graphics and animation soon become boring and annoying.
4. Download time for smartphone apps must be short.

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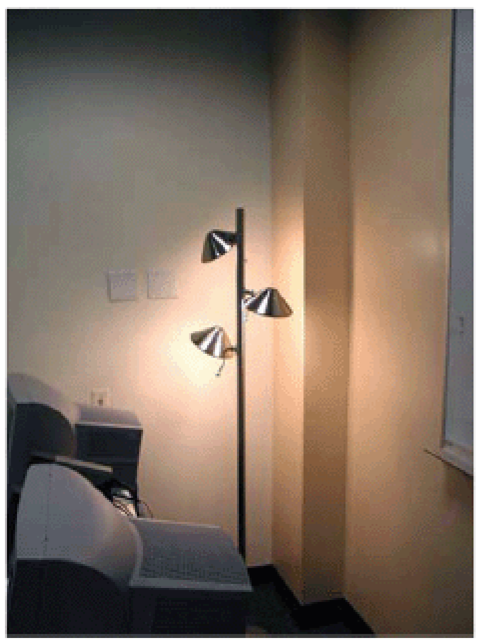
Another important issue when designing and evaluating web pages and other types of system is their accessibility to a broad range of users. In the USA, a requirement known as Section 508 of the Rehabilitation Act came into effect in 2001. The act requires that all federally funded IT systems be accessible for people with disabilities. The guidelines provided by this Act can be used as heuristics to check that systems comply with it (see Case Study 14.1). Mankoff *et al* (2005) also used guidelines as heuristics to evaluate specific kinds of usability. They discovered that developers doing a heuristic evaluation using a screen reader found 50% of known usability problems – which was more successful than user testing directly with blind users.

Heuristic evaluation has also been used to evaluate abstract aesthetic peripheral displays that portray non-critical information at the periphery of the user's attention (Mankoff *et al*, 2003). Since these devices are not designed for task performance, the researchers had to develop a set of heuristics that took this into account. They did this by developing two ambient displays: one indicated how close a bus is to the bus-stop by showing its number move upwards on a screen; the other indicated how light or dark it was outside by lightening or darkening a light display (see Figure 15.2). Then they modified Nielsen's heuristics to address the characteristics of ambient displays and asked groups of experts to evaluate the displays using them.

(a)



(b)

Figure 15.2 Two ambient devices: (a) bus indicator, (b) lightness and darkness indicator

Source: **J. Mankoff, A. K. Dey, G. Hsich, J. Kientz, Lederer and A. Morgan (2003) Heuristic evaluation of ambient devices. In *Proceedings of CHI 2003*, ACM Fig.1, p. 170. ©2003 Association for Computing Machinery, Inc. Reprinted by permission.**

The heuristics that they developed included some that were specifically geared towards ambient systems such as:

- Visibility of state: The state of the display should be clear when it is placed in the intended setting.
- Peripherality of display: The display should be unobtrusive and remain so unless it requires the user's attention. Users should be able to easily monitor the display.

In this study the researchers found that three to five evaluators were able to identify 40–60% of known usability issues. In a follow-up study, different researchers used the same heuristics with different ambient applications (Consolvo and Towle, 2005). They found 75% of known usability problems with eight evaluators and 35–55% were found with three to five evaluators, suggesting that the more evaluators you have, the more accurate the results will be – as other researchers have also reported.

Heuristics for other products, adapted from Nielsen's original heuristics, have continued to be developed and include heuristics for evaluating shared groupware (Baker *et al*, 2002), video games (Pinelle *et al*, 2008), multi-player games (Pinelle *et al*, 2009), online communities (Preece and Shneiderman, 2009), and information visualization (Forsell and Johansson, 2010). Shneiderman's eight golden rules, a set of guidelines first developed in the mid-1980s, are also frequently adapted for use with different types of systems and environments and used as heuristics for identifying usability problems (Shneiderman and Plaisant, 2010; http://www.cs.umd.edu/~ben/goldenrules.html):

1. *Strive for consistency*. Consistent sequences of actions should be required in similar situations: identical terminology should be used in prompts, menus, and help screens; and consistent color, layout, capitalization, fonts, and so on should be employed throughout. Exceptions, such as required confirmation of the delete command or no echoing of passwords, should be comprehensible and limited in number.

2. *Cater to universal usability*. Recognize the needs of diverse users and design for plasticity, facilitating transformation of content. Novice to expert differences, age ranges, disabilities, and technological diversity each enrich the spectrum of requirements that guides design. Adding features for novices, such as explanations, and features for experts, such as shortcuts and faster pacing, can enrich the interface design and improve perceived system quality.
3. *Offer informative feedback*. For every user action, there should be system feedback. For frequent and minor actions, the response can be modest, whereas for infrequent and major actions, the response should be more substantial. Visual presentation of the objects of interest provides a convenient environment for showing changes explicitly.
4. *Design dialogs to yield closure*. Sequences of actions should be organized into groups with a beginning, middle, and end. Informative feedback at the completion of a group of actions gives operators the satisfaction of accomplishment, a sense of relief, a signal to drop contingency plans from their minds, and an indicator to prepare for the next group of actions. For example, e-commerce websites move users from selecting products to the checkout, ending with a clear confirmation page that completes the transaction.
5. *Prevent errors*. As much as possible, design the system such that users cannot make serious errors; for example, gray out menu items that are not appropriate and do not allow alphabetic characters in numeric entry fields. If a user makes an error, the interface should detect the error and offer simple, constructive, and specific instructions for recovery. For example, users should not have to retype an entire name-address form if they enter an invalid zip code, but rather should be guided to repair only the faulty part. Erroneous actions should leave the system state unchanged, or the interface should give instructions about restoring the state.

6. *Permit easy reversal of actions*. As much as possible, actions should be reversible. This feature relieves anxiety, since the user knows that errors can be undone, and encourages exploration of unfamiliar options. The units of reversibility may be a single action, a data-entry task, or a complete group of actions, such as entry of a name-address block.
7. *Support internal locus of control*. Experienced users strongly desire the sense that they are in charge of the interface and that the interface responds to their actions. They don't want surprises or changes in familiar behavior, and they are annoyed by tedious data-entry sequences, difficulty in obtaining necessary information, and inability to produce their desired result.
8. *Reduce short-term memory load*. Humans' limited capacity for information processing in short-term memory (the rule of thumb is that we can remember 'seven plus or minus two chunks' of information) requires that designers avoid interfaces in which users must remember information from one screen and then use that information on another screen. It means that smartphones should not require re-entry of phone numbers, website locations should remain visible, multiple-page displays should be consolidated, and sufficient training time should be allotted for complex sequences of actions.

Doing Heuristic Evaluation

Heuristic evaluation has three stages:

1. The briefing session, in which the experts are told what to do. A prepared script is useful as a guide and to ensure each person receives the same briefing.
2. The evaluation period, in which each expert typically spends 1–2 hours independently inspecting the product, using the heuristics for guidance. The experts need to take at least two passes through the interface. The first pass gives a feel for the flow of the interaction and the product's scope. The second pass allows the evaluator to focus on specific interface elements in the context of the whole product, and to identify potential usability problems.

If the evaluation is for a functioning product, the evaluators need to have some specific user tasks in mind so that exploration is focused. Suggesting tasks may be helpful but many experts suggest their own tasks. However, this approach is less easy if the evaluation is done early in design when there are only screen mockups or a specification; the approach needs to be adapted to the evaluation circumstances. While working through the interface, specification, or mockups, a second person may record the problems identified, or the evaluator may think aloud. Alternatively, she may take notes herself. Evaluators should be encouraged to be as specific as possible and to record each problem clearly.

3. The debriefing session, in which the evaluators come together to discuss their findings and to prioritize the problems they found and suggest solutions.

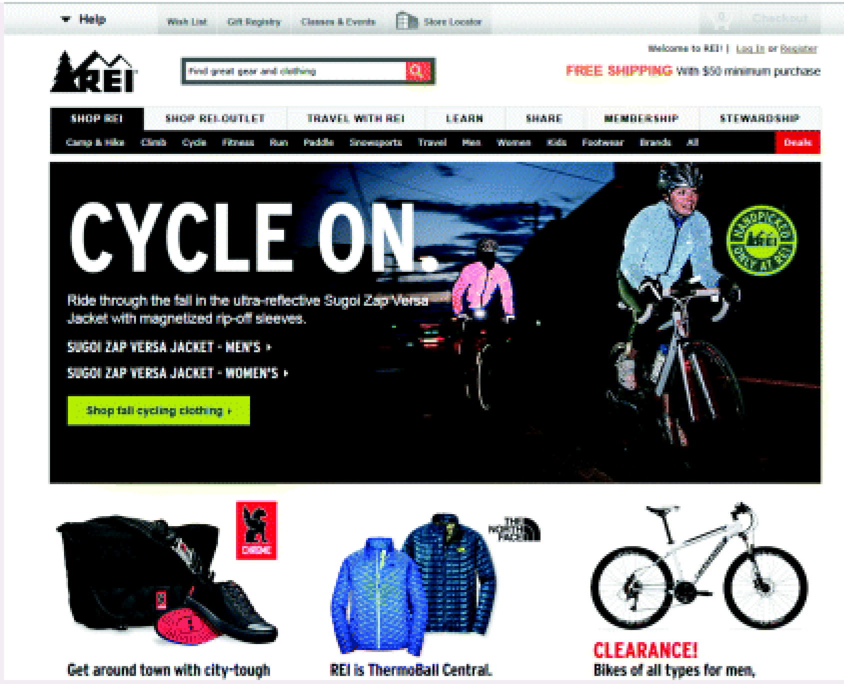
The heuristics focus the evaluators' attention on particular issues, so selecting appropriate heuristics is critically important. Even so, there is sometimes less agreement among evaluators than is desirable, as discussed in the next Dilemma.

There are fewer practical and ethical issues in heuristic evaluation than for other methods because users are not involved. A week is often cited as the time needed to train evaluators (Nielsen and Mack, 1994), but this depends on the person's initial expertise. Typical users can be taught to do heuristic evaluation, although there have been claims that this approach is not very successful (Nielsen, 1994a). A variation of this method is to take a team approach that may also involve users.

Activity 15.3

Look at the Nielsen (2014) heuristics and consider how you would use them to evaluate a website for purchasing clothes (e.g. www.rei.com, which has a homepage similar to that in Figure 15.3).

1. Do the heuristics help you focus on the website more intently than if you were not using them?
2. Might fewer heuristics be better? Which might be combined and what are the trade-offs?
3. If you did this activity using a smartphone, were all the heuristics relevant and useful?



[Figure 15.3](#) Part of the homepage of [rei.com](#)

Source: [www.rei.com](#).

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Dilemma

Classic Problems or False Alarms?

You might have the impression that heuristic evaluation is a panacea for designers, and that it can reveal all that is wrong with a design. However, it has problems. Shortly after heuristic evaluation was developed, several independent studies compared heuristic evaluation with other methods, particularly user testing. They found that the different approaches often identify different problems and that sometimes heuristic evaluation misses severe problems (Karat, 1994). This argues for using complementary methods. Furthermore, heuristic evaluation should not be thought of as a replacement for user testing.

Another problem concerns experts reporting problems that don't exist. In other words, some of the experts' predictions are wrong (Bailey, 2001). Bailey cites analyses from three published sources showing that only around 33% of the problems reported were real usability problems, some of which were serious, others trivial. However, the heuristic evaluators missed about 21% of users' problems. Furthermore, about 43% of the problems identified by the experts were not problems at all; they were false alarms! Bailey points out that this means only about half the problems identified are true problems: "More specifically, for every true usability problem identified, there will be a little over one false alarm (1.2) and about one half of one missed problem (0.6). If this analysis is true, heuristic evaluators tend to identify more false alarms and miss more problems than they have true hits."

How can the number of false alarms or missed serious problems be reduced? Checking that experts really have the expertise that they claim would help, but how can this be done? One way to overcome these problems is to have several evaluators. This helps to reduce the impact of one person's experience or poor performance. Using heuristic evaluation along with user testing and other methods is also a good idea. ■

While heuristic evaluation has proven to be very useful and used by many usability experts over the years, the guidelines were designed to be applied to PC-based applications that were being developed in the 1980s. Given the range of current technologies (e.g. smartphones and various other mobile devices, tablets, tabletops, and ambient systems) that have been developed since Nielsen devised his original set of heuristics – it raises the question of whether they can continue to be of use to test the usability and user experience of new products. For example, how useful will they be for evaluating wearable technologies, such as smartwatches and augmented glasses? What other heuristics might be needed? There is also overlap between how evaluation heuristics and other kinds of design guidelines are used. Design guidelines are often converted into heuristics so it is possible to use these instead or in combination with heuristic evaluation.

15.2.2 Walkthroughs

Walkthroughs have been used by system developers for many years for inspecting code and the concept offers an alternative approach to heuristic evaluation for predicting users' problems without doing user testing. As the name suggests, walkthroughs involve walking through a task with the product and noting problematic usability features. Most walkthrough methods do not involve users. Others, such as pluralistic walkthroughs, involve a team that may include users, as well as developers, and usability specialists.

In this section we consider cognitive and pluralistic walkthroughs. Both were originally developed for desktop systems but, as with heuristic evaluation, they can be adapted to web-based systems, smartphones and mobile devices, tabletops, and products such as DVD players.

Cognitive Walkthroughs

"Cognitive walkthroughs involve simulating a user's problem-solving process at each step in the human–computer dialog, checking to see if the user's goals and memory for actions can be assumed to lead to the next correct action" (Nielsen and Mack, 1994, p. 6). The defining feature is that they focus on evaluating designs for ease of learning – a focus that is motivated by observations that users learn by exploration (Wharton *et al*, 1994). The steps involved in cognitive walkthroughs are:

1. The characteristics of typical users are identified and documented and sample tasks are developed that focus on the aspects of the design to be evaluated. A description, mockup, or prototype of the interface to be developed is also produced, along with a clear sequence of the actions needed for the users to complete the task.
2. A designer and one or more expert evaluators come together to do the analysis.
3. The evaluators walk through the action sequences for each task, placing it within the context of a typical scenario, and as they do this they try to answer the following questions:
   - Will the correct action be sufficiently evident to the user? (Will the user know what to do to achieve the task?)
   - Will the user notice that the correct action is available? (Can users see the button or menu item that they should use for the next action? Is it apparent when it is needed?)
   - Will the user associate and interpret the response from the action correctly? (Will users know from the feedback that they have made a correct or incorrect choice of action?)

   In other words: will users know what to do, see how to do it, and understand from feedback whether the action was correct or not?

4. As the walkthrough is being done, a record of critical information is compiled in which:
    - The assumptions about what would cause problems and why are identified.
    - Notes about side issues and design changes are made.
    - A summary of the results is compiled.
5. The design is then revised to fix the problems presented.

    As with heuristic and other evaluation methods, developers and researchers can modify the method to meet their own needs more closely, or adapt the process so that it applies to particular features of mobile or other new technologies.

When doing a cognitive walkthrough it is important to document the process, keeping account of what works and what doesn't. A standardized feedback form can be used in which answers are recorded to each question. Any negative answers are carefully documented on a separate form, along with details of the system, its version number, the date of the evaluation, and the evaluators' names. It is also useful to document the severity of the problems: for example, how likely a problem is to occur and how serious it will be for users. The form can also record the process details outlined in points 1 to 4 as well as the date of the evaluation.

Compared with heuristic evaluation, this technique focuses more closely on identifying specific user problems at a high level of detail. Hence, it has a narrow focus that is useful for focusing on certain types of system rather than for evaluating whole systems. In particular, it can be useful for applications involving complex cognitive operations. However, it is very time-consuming and laborious to do and evaluators need a good understanding of the cognitive processes involved.

Activity 15.4

Conduct a cognitive walkthrough to find a copy of this book to buy as an ebook at www.amazon.com or www.wiley.com.

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Activity 15.5

Activity 15.3 asked you to do a heuristic evaluation of www.rei.com or a similar online retail site. Now go back to that site and do a cognitive walkthrough to buy some clothing. When you have completed the evaluation, compare your findings from the cognitive walkthrough with those from heuristic evaluation.

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Another variation of cognitive walkthrough was developed by Rick Spencer of Microsoft, to overcome some problems that he encountered when using the original form of cognitive walkthrough (Spencer, 2000). The first problem was that answering the three questions in step 3 and discussing the answers took too long. Second, designers tended to be defensive, often invoking long explanations of cognitive theory to justify their designs. This second problem was particularly difficult because it undermined the efficacy of the method and the social relationships of team members. In order to cope with these problems, Rick Spencer adapted the method by reducing the number of questions and curtailing discussion. This meant that the analysis was more coarse-grained but could be completed in about 2.5 hours. He also identified a leader, the usability specialist, and set strong ground rules for the session, including a ban on defending a design, debating cognitive theory, or doing designs on the fly.

These adaptations made the method more usable, despite losing some of the detail from the analysis. Perhaps most important of all, Spencer directed the social interactions of the design team so that they achieved their goals.

Link to an up-to-date discussion on the value of the cognitive walkthrough method for evaluating various devices, at http://www.userfocus.co.uk/articles/cogwalk.html

Pluralistic Walkthroughs

“Pluralistic walkthroughs are another type of walkthrough in which users, developers and usability experts work together to step through a [task] scenario, discussing usability issues associated with dialog elements involved in the scenario steps” (Nielsen and Mack, 1994, p. 5). In a pluralistic walkthrough, each of the evaluators is asked to assume the role of a typical user. Scenarios of use, consisting of a few prototype screens, are given to each evaluator who writes down the sequence of actions they would take to move from one screen to another, without conferring with fellow panelists. Then the panelists discuss the actions they each suggested before moving on to the next round of screens. This process continues until all the scenarios have been evaluated (Bias, 1994).

The benefits of pluralistic walkthroughs include a strong focus on users’ tasks at a detailed level, i.e. looking at the steps taken. This level of analysis can be invaluable for certain kinds of systems, such as safety-critical ones, where a usability problem identified for a single step could be critical to its safety or efficiency. The approach lends itself well to participatory design practices by involving a multidisciplinary team in which users play a key role. Furthermore, the group brings a variety of expertise and opinions for interpreting each stage of an interaction. Limitations include having to get all the experts together at once and then proceed at the rate of the slowest.

Furthermore, only a limited number of scenarios, and hence paths through the interface, can usually be explored because of time constraints.

15.3 Analytics

Analytics is a method for evaluating user traffic through a system (discussed in Chapters 7, 8, and 13) – mostly such systems involve selling a product or service so knowing what customers do and want is imporant for improving the product or service, but analytics are also being used for evaluating non-transactional systems (e.g. Sleeper *et al*, 2014). When used to examine traffic on a website or part of a website as mentioned in Chapter 7, the analytics are known as web analytics. Web analytics can be collected locally or remotely across the Internet by logging user activity, counting and analyzing the data in order to understand what parts of the website are being used and when. Although analytics are a form of evaluation that is particularly useful for evaluating the usability of a website, they are also valuable for business planning including the business of online education. Recently, analytics have been applied to understand how learners in massive open online courses (MOOCs) and other web-based education courses interact with these systems (e.g. Oviatt *et al*, 2013). For example, the developers of these systems are interested in such questions as: Why do learners enroll in these systems and at what point do learners tend to drop out and why? What are the characteristics of learners who complete the course compared with those who do not complete it?

> Video of Simon Buckingham Shum's 2014 keynote presentation at the EdMedia 2014 Conference which provides an introduction to learning analytics, and how analytics are used to answer key questions in a world where people are dealing with large volumes of digital data — see it at
>
> http://people.kmi.open.ac.uk/sbs/2014/06/edmedia2014-keynote/

Many companies are developing their own analytics tools; others use the services of companies such as Google and VisiStat, which specialize in providing analytics and the analysis necessary to understand large volumes of data.Typically these companies present their analysis in forms that are easy for developers and website managers to understand – e.g. as graphs, tables, and other types of data visualizations. The home page of VisiStat (VisiStat.com) contains a short promotional video describing the kinds of services that the company offers and why these services are important to businesses. An example of how web analytics can be used to analyze and help developers to improve website performance is provided by VisiStat's analysis of Mountain Wines' website (VisiStat, 2010).

Mountain Wines, located in Saratoga, California, aims to create memorable experiences for guests who visit the vineyard. Following on the tradition started by Paul Masson, world famous wine maker, Mountain Wines offers a beautiful venue for a variety of events including weddings, corporate meetings, dinner parties, birthdays, concerts, and vacations. Mountain Wines uses a variety of advertising media to attract customers to its website and in 2010 invested about $10 000 a month in advertising. However, the website has remained unchanged for several years because the company didn't have a way of evaluating its effectiveness and could not decide whether to increase or decrease investment in it. They then decided to employ VisiStat, to use their web analytics tool. Prior to enlisting this company, the only record that Mountain Wines had of the effectiveness of its advertising came from its front-desk employees who were instructed to ask visitors 'How did you hear about Mountain Wines?'

VisiStat provided Mountain Wines with data showing how their website was being used by potential customers, e.g. data like that shown in Figures 15.4 to 15.6. Figure 15.4 provides an overview of the number of page views of the website per day. Figure 15.5 provides additional details and shows the hour-by-hour traffic. Clicking on the first icon for more detail shows where the IP addresses of the traffic are located (Figure 15.6). VisiStat can also provide information about such things as which visitors are new to the site, which are returners, and which other pages visitors came from.

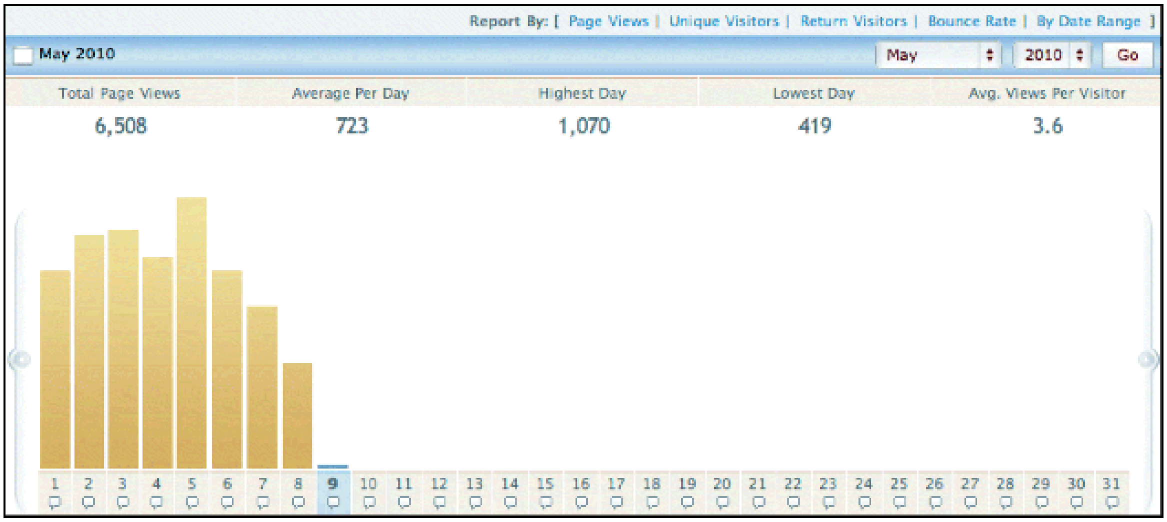

Figure 15.4 A general view of the kind of data provided by VisiStat

Source: http://www.visistat.com/tracking/monthly-page-views.php

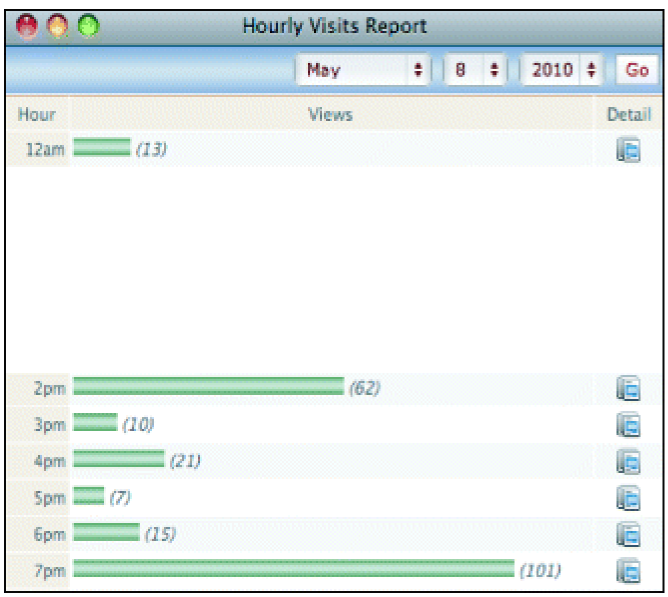

Figure 15.5 Clicking on May 8 provides an hourly report from midnight until 10.00 p.m. (only midnight and 2.00 p.m.–7.00 p.m. shown)

Source: http://www.visistat.com/tracking/monthly-page-views.php

Display By: Geographic Location

	Unique Visitor	Views	Detail
1.	Los Angeles, California	6	
2.	Sharpsburg, Maryland	1	
3.	Phoenix, Arizona	3	
4.	Lemesos, Limassol	2	
5.	Targu-mures, Mures	1	

Figure 15.6 Clicking on the icon for the first hour in Figure 15.5 shows where the IP addresses of the 13 visitors to the website are located

Source: http://www.visistat.com/tracking/monthly-page-views.php

Using this data and other data provided by VisiStat, Mountain Wines could see visitor totals, traffic averages, traffic sources, visitor activity, and more. They discovered the importance of visibility for their top search words; they could pinpoint where guests were going on their website; and they could see where their guests were geographically located.

Activity 15.6

1. How were users involved in the Mountain Wines website evaluation?
2. From the information described above, how might Mountain Wines have used the results of the analysis to improve its website?
3. Where was the evaluation carried out?

Comment

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More recently, other types of specialist analytics have also been developed such as visual analytics, in which thousands and sometimes millions of data points are displayed and manipulated visually, such as Hansen *et al*'s (2011) social network analysis (see Figure 15.7).

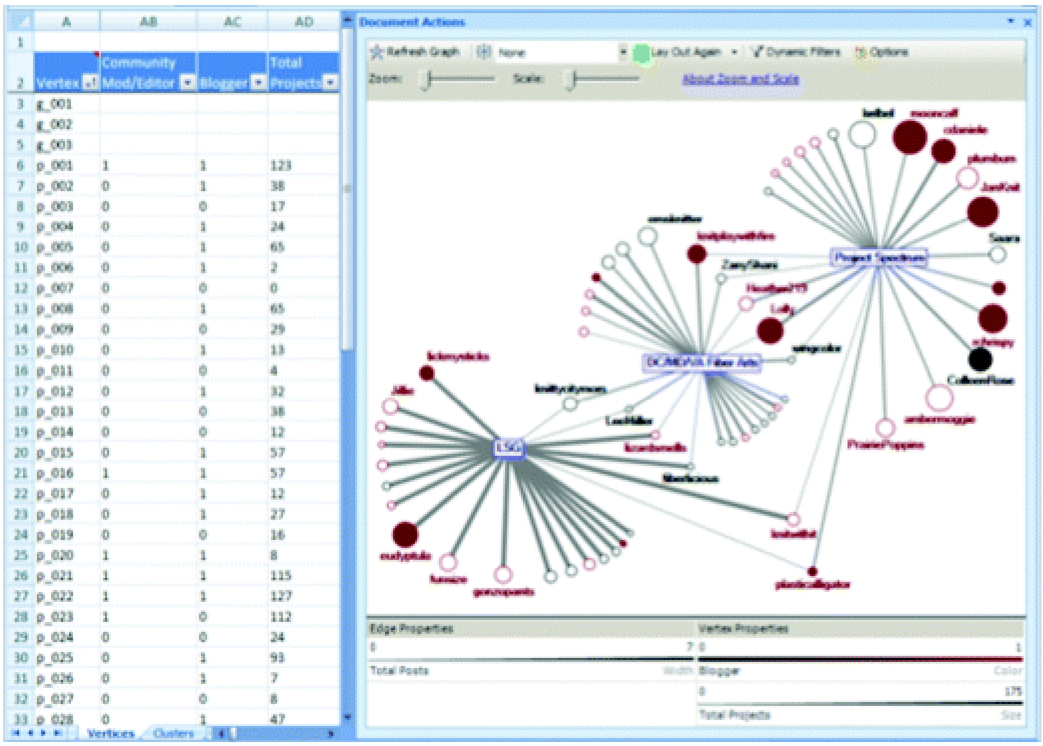

Figure 15.7 Social network analysis showing clusters and the relationships between the entities and the clusters

Source: **Perer, A. and Shneiderman, B. (2008) Integrating Statistics and Visualization: Case Studies of Gaining Clarity during Exploratory Data Analysis. *CHI 2008 Proceedings*. Visual Synthesis April 5–10, 2008, 265–274. ©2008 Association for Computing Machinery, Inc. Reprinted by permission.**

Lifelogging is another interesting variation that can be used for evaluation as well as for sharing information with friends, family, and colleagues. Typically, lifelogging involves recording GPS location data and personal interaction data on smartphones. In an evaluation context this can raise privacy concerns. Even though users tend to get used to being logged, they generally want to remain in control of the logging (Kärkkäinen *et al*, 2010).

Dilemma

Analyzing Workers' Social Networking Behavior – an Invasion of Privacy?

Some analytics software can be used by IT administrators to track workers' behavior on social networking sites during working hours. The data collected can be used to determine who is collaborating with whom, and to inform developers about how much their applications are being used – a concept often referred to as stickiness. While these reasons for tracking users appear to be bona fide, is this a threat to personal privacy? ■

15.4 Predictive Models

Similar to inspection methods and analytics, predictive models evaluate a system without users being present. Rather than involving expert evaluators role-playing users as in inspections, or tracking their behavior as in analytics, predictive models use formulas to derive various measures of user performance. Predictive modeling provides estimates of the efficiency of different systems for various kinds of task. For example, a smartphone designer might choose to use a predictive model because it can enable her to determine accurately which is the optimal layout of keys on the phone for allowing common operations to be performed.

Two types of predictive models have been particularly influential in HCI and interaction design over the years: Fitts' Law, which we discuss in more detail below, and the GOMS family of models (Card *et al*, 1983), which are rarely used now as the tools were designed in the early 1980s, primarily for comparing PC-based interfaces for tasks that were highly predictable (now archived from the 3rd edition of this book on the ID-Book.com website).

15.4.1 Fitts' Law

Fitts' Law (Fitts, 1954) predicts the time it takes to reach a target using a pointing device. It was originally used in human factors research to model the relationship between speed and accuracy when moving towards a target on a display. In interaction design, it has been used to describe the time it takes to point at a target, based on the size of the object and the distance to the object. Specifically, it is used to model the time it takes to select objects on a screen, such as alphanumerics, for different spatial configurations. One of its main benefits is that it can help designers decide where to locate physical or digital buttons, what size they should be, and how close together they should be on a touch display or a physical device. To begin with, it was most useful for designing physical laptop/PC keyboard layouts and the placement of physical keys on mobile devices, such as phones, watches, and remote controls. Since then, it has been used more for designing the layout of digital input displays for touch screen interfaces, especially, for considering the effectiveness of new ways of single-digit typing such as using finger swiping (see Chapter 6).

Fitts' Law states that:

$$T = k \log_2(D/S + 1.0)$$

where

T = time to move the pointer to a target

D = distance between the pointer and the target

S = size of the target

k is a constant of approximately 200 ms/bit

In a nutshell, the bigger the target, the easier and quicker it is to reach it. This is why interfaces that have big buttons are easier to use than interfaces that present lots of tiny buttons crammed together. Fitts' Law also predicts that the most quickly accessed targets on any computer display are the four corners of the screen. This is because of their pinning action, i.e. the sides of the display constrain the user from over-stepping the target. However, as pointed out by Tog on his AskTog.com website, corners seem strangely to be avoided at all costs by designers.

Fitts' Law can be useful for evaluating systems where the time to physically locate an object is critical to the task at hand. In particular, it can help designers think about where to locate objects on the screen in relation to each other. This is especially useful for mobile devices, where there is limited space for placing icons and buttons on the screen. For example, in a study carried out by Nokia, Fitts' Law was used to predict expert text entry rates for several input methods on a 12-key cell phone keypad (Silverberg *et al*, 2000). The study helped the designers make decisions about the size of keys, their positioning, and the sequences of presses to perform common tasks. Trade-offs between the size of a device and accuracy of using it were made with the help of calculations from this model. Fitts' Law has also been used to compare eye-tracking input with manual input for visual targets (Vertegaal, 2008); to compare different ways of mapping Chinese characters to the keypad of cell phones (Liu and Räihä, 2010); to investigate the effect of the size of the physical gap between displays and the proximity of targets in multiple-display environments (Hutchings, 2012); and to evaluate tilt as an input method for devices with built-in accelerometers, such as touch screen phones and tablet computers (MacKenzie and Teather, 2012). The AskTog.com website also discusses the application of Fitts' Law to smartphones and other new technologies with limited screen real estate.

Activity 15.7

Microsoft toolbars provide the user with the option of displaying a label below each tool. Give a reason why labeled tools may be accessed faster. (Assume that the user knows the tool and does not need the label to identify it.)

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Assignment

This assignment continues the work you did on the new interactive product for booking tickets online at the end of Chapters 10, 11, and 14. The aim of this assignment is to evaluate the prototypes produced in the assignment of Chapter 11 using heuristic evaluation.

a. Decide on an appropriate set of heuristics and perform a heuristic evaluation of one of the prototypes you designed in Chapter 11.

b. Based on this evaluation, redesign the prototype to overcome the problems you encountered.

c. Compare the findings from this evaluation with those from the usability testing in the previous chapter. What differences do you observe? Which evaluation approach do you prefer and why?

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www.id-book.com/quickvotes/chapter15

Summary

This chapter presented inspection evaluation methods, focusing on heuristic evaluation and walkthroughs which are usually done by specialists (usually referred to as experts), who role-play users' interactions with designs, prototypes, and specifications using their substantial knowledge of the kinds of problems that users typically encounter and then offer their opinions. Heuristic evaluation and walkthroughs offer the evaluator a structure to guide the evaluation process.

Analytics, in which user interaction are logged, are often performed remotely and without users being aware that their interactions are being tracked. Very large volumes of data are collected, anonymized, and statistically analyzed using specially developed software services. The analysis provides information about how a system is used, e.g. how different versions of a website or prototype perform, or which parts of a website are seldom used – possibly due to poor usability design or lack of appeal. Data are often presented visually so that it is easier to see trends and interpret the results.

Fitts' Law is an example of a technique that can be used to predict user performance by determining whether a proposed interface, system, or keypad layout will be optimal. Typically Fitts' Law is used to compare different design layouts for virtual or physical objects, such as buttons on a device or screen, for a small sequence of frequently performed tasks.

Evaluators typically find that they have to modify this method for evaluations with the wide range of products that have come onto the market since Fitts' Law was originally developed; this is similar to the other methods discussed in earlier chapters, which frequently also have to be modified.

Key points

- Inspections can be used for evaluating a range of representations including requirements, mockups, functional prototypes, or systems.
- User testing and heuristic evaluation often reveal different usability problems.
- Other types of inspections used in interaction design include pluralistic and cognitive walkthroughs.
- Walkthroughs are very focused and so are suitable for evaluating small parts of a product.
- Analytics involves collecting data about the interactions of users in order to identify which parts of a website or prototype are underused.
- When applied to websites, analytics are often referred to as ‘web analytics.’ Similarly, when applied to learning systems, they are referred to as ‘learning analytics.'
- Fitts’ Law is an example of a predictive model that can be used to predict performance involving the selection of virtual or physical objects.
- Predictive models require neither users nor usability experts to be present, but the evaluators must be skilled in applying the models.
- Predictive models like Fitts’ Law are used to evaluate systems with limited, clearly defined, functionality such as data entry applications, and key-press sequences for smartphones and other handheld devices.

Further Reading

BUDIU, R. and NIELSEN, J. (2012) *Mobile Usability*. New Riders Press. This book discusses why designing for mobile devices is different from designing for other systems. It describes how to evaluate these systems, including doing expert reviews, and provides many examples.

MacKenzie, I. S. (1992) Fitts' Law as a research and design tool in human–computer interaction. *Human–Computer Interaction, 7*, 91–139. This early paper by Scott Mackenzie, an expert in the use of Fitts' Law, provides a detailed discussion of how it can be used in HCI.

Mackenzie, I. S. and Soukoreff, R. W. (2002) Text entry for mobile computing: models and methods, theory and practice. *Human–Computer Interaction, 17*, 147–198. This paper provides a useful survey of mobile text-entry techniques and discusses how Fitts' Law can inform their design.

Mankoff, J., Dey, A. K., Hsich, G., Kientz, J. and Lederer, M. A. (2003) Heuristic evaluation of ambient devices. *Proceedings of CHI 2003, ACM, 5*(1), 169–176. More recent papers are available on this topic but we recommend this paper because it describes how to derive rigorous heuristics for new kinds of applications. It illustrates how different heuristics are needed for different applications.